Two elements of a Sigma type over propositions with the same first component are equal

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 467-470 and its proof environment opens on line 471, the next line; the pairing is by adjacency and the retained proof range 471-475 is the whole environment.

Editorial note. The retained counter values (chapter 3, section 5, theorem offset 0) make the counted lemma print as Lemma 3.5.1, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:path-subset = 3.5.1).

Editorial note. One label the proof cites lies outside the card, thm:path-sigma (book Theorem 2.7.2, whose chapter file is not part of the pinned corpus so the pointer span is the label line of this chapter dump), and is declared in external_references as a non-load-bearing navigation pointer: the retained proof states the criterion it applies.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the path criterion for Sigma types over mere propositions (author label thm:path-subset, book Lemma 3.5.1).

Allowed standard prerequisites (editorial).

- Sigma type over a family of mere propositions. The type consists of a point of the base together with an element of a family that is assumed to be a family of mere propositions.. It is the type whose elements are compared.
- Paths in a Sigma type. A path between two pairs is given by a path between the first components together with a path between the transported second component and the second component of the target.. It is the criterion that reduces the goal to one equation.
- Mere proposition. Any two elements of a mere proposition are equal.. It closes the remaining equation.

Source reference. Original theorem 2.7.2: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 1853–1853. Book Theorem 2.7.2, the path criterion

for Sigma types, applied at source line 473. Navigation pointer only: the chapter file that carries it is not part of the pinned corpus, so the pointer span is the label line of this chapter dump.

Lemma 3.5.1. *Suppose $P : A \to \mathcal{U}$ is a type family such that $P(x)$ is a mere proposition for all $x : A$. If $u, v : \sum_{(x:A)} P(x)$ are such that $\mathsf{pr}_1(u) = \mathsf{pr}_1(v)$, then $u = v$.*

Proof. Suppose $p : \mathsf{pr}_1(u) = \mathsf{pr}_1(v)$. By original theorem 2.7.2, to show $u = v$ it suffices to show $p_*(\mathsf{pr}_2(u)) = \mathsf{pr}_2(v)$. But $p_*(\mathsf{pr}_2(u))$ and $\mathsf{pr}_2(v)$ are both elements of $P(\mathsf{pr}_1(v))$, which is a mere proposition; hence they are equal. □